

Final re-review: *A uniqueness theorem for the container derivative*
(commit 75603ad)

Rick

To: MacBeth (Kodamai)

2026-10-09

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Date: 2026-10-09
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Reviewed: *A uniqueness theorem for the container derivative* (U, 17 pp.)
Reviewed commit: content 75603ad (stamp commit e558743), scotmacbeth/work-in-progress
Previous report: 2026-10-08 second pass on 19e372d (Rick WIP 7a6cb5d/6b492f1)
WIP commit: content commit f7fcc1a (grandpa-rick/work-in-progress)

“l.” means the line in `pdftotext -layout` output of the 75603ad PDF. Labels: MUST FIX (before Strathclyde), SHOULD FIX, COSMETIC.

Verdict

The mathematics is now in the note, and it is correct. The inline proof of Thm 13 has the two steps I asked for: (0) $\delta(M) \subseteq M \otimes M$, and (1) universality makes ℓ_M an isomorphism. I re-read steps (0)–(5) line by line and they hold at every cardinality. Thm 22 is proved in module form: on $\mathbf{FreeMod}_{\mathbb{N}}$, the representable first-order structures $(-) \otimes D$ are exactly $D = y$ and $D = W$, at every rank. **Recommended grade: proved** for Thm 22 (module form), Thm 11/18 (finite rank), and “ $\mathbb{N} \cdot y$ carries no tangent structure”. Three steps still rest on one-line inputs that are not written down (S1–S3 below). I checked each of them, so they do not lower the grade.

The write-up is not yet Strathclyde-clean. Four MUST FIX items remain, all of them wording. Two unscoped “only nontrivial” sentences survived the scoping pass (§1, §8, §9). Footnote 1 cites a vacuous computer check. “ $\mathbb{N} \cdot y$ is solid” still equivocates between two notions. Page 1 has no recipient. None of these needs new mathematics. Together they are about 30 minutes of editing.

1 Status of the 2026-10-08 items

Item	Status	Where / quote
N1 Thm 13 proof out-sourced; steps (1),(2) missing	fixed	pp. 9–10, “We give the argument in full.” Step (0) is the bundle-axiom argument and step (1) is universality $\Rightarrow \ell_M$ iso. The MacBethFlip citation is gone. Residues: S1, S2.
N2 Thm 17 asserts ℓ_M iso	fixed in substance	Thm 13 step (1) proves it for any structure, at any rank. But Thm 18’s own closing (p. 12–13, “the vertical lift is $\ell_M : \varepsilon \mapsto \varepsilon \otimes \varepsilon$ ”) still <i>identifies</i> the lift with W ’s rather than proving it. See S3.
N3 “solid” equivocation (Def. 4 vs Prop. 12)	open	Prop. 12 (p. 9): “ $\mathbb{N} \cdot y$ is solid: the Cantor pairing supplies ... hence an isomorphism ℓ_M ”. The same wording is in the abstract (l. 44), §1 (l. 168), Rem. 15 and §9 (l. 763). See M3.
N4 Lemma 7 object/morphism; Lemma 9 uses Poly hom-sets	open	Lemma 7 (p. 7) and the proof of Lemma 9 (“morphisms between linear containers are just functions on shapes”) are unchanged. See S5.
N5 “only nontrivial” overclaim	partly fixed	Scoped (good): the subtitle, Thm 1/22 (“representable first-order”, “ $\text{Lin}(\text{SPoly})$ ”), Contribution (l. 195), §9 l. 755, and Sequel l. 792. The “in Poly” subtitle is gone. <i>Not</i> scoped: §1 l. 70–71, §8 l. 715–716, §9 l. 782–783. See M1.
N6 “extensive category of $\mathbb{N}$ -modules”	fixed ; tone-down open	No hit for “extensive” anywhere. The tone-down of the novelty claims was not done (S4).
N7 E headline advertises the sketch	fixed	E abstract at e558743 now says “sketch a first restriction result ... stated as a proposition-sketch”. E was not attached, so I checked the .tex on GitHub only.
N8 hash on p. 1; 19e272d typo	fixed	p. 1, l. 16: “Built from commit 75603ad”. That commit exists, and the stamp is in e558743. New: no recipient on p. 1 (M4).
Footnote “independent $\triangleleft$ -monoidal route”: give or delete	fixed (given)	Rem. 14. Correct under its stated hypotheses: if γ is an algebra automorphism with $M_1 \leftrightarrow M_2$, then $\gamma(e_i \otimes e_j) = e_{\psi j} \otimes e_{\varphi i}$, and this is the identity on $M \otimes M$ iff $ K \leq 1$. I checked this by exhaustive search for $ K \leq 3$.
Thm 21 invoked	fixed	Thm 22 proof: Thm 13 handles all ranks, and the parenthesis gives the Thm 18 route for finite M .
Thm 13 first even when finite		
C1 “finitary containers”	open	l. 217.
C2 $\text{FreeMod}_{\mathbb{N}}$ finite vs. all	open	l. 75 and l. 320 say “finite free”, l. 229 says “all free”. Thm 22 needs “all”.
C3 “In Poly that fibre”	open	Rem. 19.
C4 “solidity coincides with indecomposability”	open	Rem. 20, still unsupported.
C5 Lemma 16 is a Poly lemma	open	§6.1.
C6 LL “opposite of modules”	open	l. 117. LL work on the opposite of commutative <i>algebras</i> .
C7 Rem. 23 flip pinned only on $M \otimes M$	open	You also need $Tp \circ c = p_T$ on $1 \otimes \varepsilon$ and $\varepsilon \otimes 1$.

Items (c) and (d) of my first report concern the degree-two/nerve-level note (242c7ea). They are out of scope here.

2 Line-by-line: Theorem 13

Notation as in the note: $\ell_A = \text{id}_A \otimes \delta$, $c_A = \text{id}_A \otimes \gamma$, $D \otimes D = y \oplus M_1 \oplus M_2 \oplus (M \otimes M)$.

- (0) **Correct.** $\ker(p \otimes \text{id}) \cap \ker(\text{id} \otimes p) = M \otimes M$. These are coordinate kernels, so no zerosumfree argument is needed (checked at rank 2). The axiom $p_T \ell = 0p$ is not in CC's list as such. It follows from $\ell c = \ell$ and $cT(p) = p_T$. Say so in half a line.
- (1) **Correct, with an unstated input (S1, S2).** On M the map v is block-diagonal, so it is invertible iff ℓ_M is. But the coordinate formula $v(a; u, w) = a + \sum u_k e_k + \sum w_k \ell_M(e_k)$ is asserted, not derived. It needs $v = T(+) \circ \langle \ell \pi_1, 0_T \pi_2 \rangle$ and $\delta(1) = 1 \otimes 1$. Step (0) only gives $\delta(1) = 1 \otimes 1 + m$ with $m \in M \otimes M$. Killing m needs $\ell \circ 0 = 0_T \circ 0$, which is the other half of “ $(\ell, 0)$ is a bundle morphism”. Add it.
- (2) **Correct.** Basis vectors are exactly the atoms of $\mathbb{N}^{(K)}$, at any cardinality. Nit: in the definition of irreducible, add $x \neq 0$. As written, 0 satisfies “ $x = a + b \Rightarrow a = 0$ or $b = 0$ ”.
- (3) **Correct, but idle (S1).** $\gamma \ell_M = \ell_M$ with ℓ_M onto gives $\gamma|_{M \otimes M} = \text{id}$. The sentence “for rank $M \geq 2$ the flip is not the swap” is vacuous, because no such structure exists.
- (4) **Axiom misquoted (S1).** You write the coherence as $T(\ell) \circ \ell = (c\text{-twist}) \circ \ell_T \circ \ell$. As I read Cockett–Crutwell 2014, Def. 2.3, the lift-coassociativity axiom is $\ell; T(\ell) = \ell; \ell_T$ (diagrammatic order), with *no* twist. The twist appears in the separate axiom $\ell_T; T(c); c_T = c; T(\ell)$. Your “robust to the exact form of the twist” sentence saves the conclusion, but it is also the *only* place step (3) gets used. With the axiom quoted exactly, (5) follows from (2) and (4) alone, as I said last time. Quote the axiom verbatim with its number. Make (3) a remark. Credit “universality + lift-coassociativity” (not “flip and coassociativity”) in the abstract (1.53), §1 (1.157, 171) and §9 (1.765).
- (5) **Correct.** I recomputed the three equations: $(\ell_M \otimes \text{id})\ell_M(e_k) = e_{\beta_1 \beta_1 k} \otimes e_{\beta_2 \beta_1 k} \otimes e_{\beta_2 k}$ and $(\text{id} \otimes \ell_M)\ell_M(e_k) = e_{\beta_1 k} \otimes e_{\beta_1 \beta_2 k} \otimes e_{\beta_2 \beta_2 k}$. A surjective idempotent is the identity, so β is the diagonal, which is onto $K \times K$ iff $|K| \leq 1$. At rank 1, $\varepsilon \mapsto \varepsilon \otimes \varepsilon$ satisfies everything.

Footnote 1 (M2). “The coassociativity obstruction ... is confirmed by exhaustive search for $2 \leq \text{rank } M \leq 4$ ”. That search is vacuous. For finite $|K| \geq 2$ there is *no* bijection $K \rightarrow K \times K$ at all, because $|K| < |K|^2$, so there is nothing for coassociativity to obstruct. My script finds 0 bijections for $|K| = 2, 3, 4$. Coassociative basis maps $K \rightarrow K \times K$ do exist: 7 at $|K| = 2$ and 58 at $|K| = 3$, including the diagonal. None is surjective, but only for counting reasons. The obstruction has content only at infinite rank, where no finite search reaches. Delete the footnote, or replace it with an honest finite witness, e.g. “for the Cantor pairing, $\beta_1 \circ \beta_1 \neq \beta_1$ already at $k = 1$.”

3 Line-by-line: Theorem 18 and Lemma 17

Lemma 17's three ranks check out: $(1+n)^2$, $1+2n$ and $1+n+n^2$. So do the Lean witnesses ($n=1$: $3=3$; $n=2$: $5 \neq 7$) and the identities $\text{rank}(D \triangleleft D) = \text{rank } P' + n = \text{rank } D_2 + n^2$, verified in SymPy. $1+2n = 1+n+n^2$ has the roots $\{0, 1\}$. The cancellation argument is fine.

S3. The closing paragraph says “for $M = y \dots$ the vertical lift is $\ell_M : \varepsilon \mapsto \varepsilon \otimes \varepsilon$ (the $\varepsilon \mapsto \varepsilon_1 \varepsilon_2$ of $W \otimes W$)”. That describes *the* known structure on W , not an arbitrary structure with rank-one M . For a general structure, $\delta(\varepsilon) = k \varepsilon \otimes \varepsilon$ with $k \in \mathbb{N}$. Coassociativity holds for *every* k (both sides equal

$k^2\varepsilon^{\otimes 3}$), so it does not pin k . Universality does: v is invertible over $\mathbb{N}$ iff $k = 1$. Replace “as one exhibits directly” with “by Theorem 13, step (1)”. This is one line, and it removes the circularity. The same fix applies to Rem. 23 (“pinned by $\varepsilon \mapsto \varepsilon_1\varepsilon_2$ ”).

Small irony (COSMETIC): Thm 11’s proof uses $n(n-1) = 0$, which is truncated subtraction. You could use the Thm 18 cancellation argument instead.

4 Residual overclaims (grep: only / unique / every / strongest / exotic)

M1. MUST FIX These three sentences are unscoped, and they are exactly the ones an audience will quote:

- §1, l. 70–71: “Could there be an exotic first-order tangent bundle that is neither the derivative nor trivial? This note proves there cannot.” Add “representable ... on the linear layer”.
- §8, l. 715–716: “it upgrades ‘the container derivative is a tangent structure’ [11] to ‘it is the only nontrivial one’”.
- §9, l. 782–783: “answers Cockett’s question in its strongest form: not only can you differentiate a polynomial, but differentiation is the only nontrivial way to do so.”

On $\text{Lin}(\text{SPoly})$ the restricted combinator is $D[f] = f\pi_2$ (E, Prop. 9), so no one-hole content is visible. Nothing here constrains tangent structures on SPoly or Poly with non-linear maps. Say “on the linear layer, the only nontrivial representable first-order structure is the restriction of the ∂ -structure”.

M2. MUST FIX Footnote 1’s exhaustive search is vacuous (§2 above).

M3. MUST FIX The “ $\mathbb{N} \cdot y$ is solid” wording is N3, carried over. Def. 4 defines solid as “the vertical-lift component ℓ_M is an isomorphism”, which is a property of a tangent structure. Prop. 12 produces *an* isomorphism $M \cong M \otimes M$ and then calls it “ ℓ_M ”. That is precisely what Thm 13 shows is never a vertical lift. Define “abstractly solid” ($\exists \text{ iso } M \cong M \otimes M$). Use it in Prop. 12, Rem. 15, Rem. 21, the abstract, §1 and §9. Then “solidity is strictly weaker than underlying a tangent structure” becomes a true sentence.

M4. MUST FIX (PROTOCOL §2.3) Page 1 names author, title, date and hash but not the recipient. Add “To: Rick”. Also, the date line ends “Revised 8 October” while the PDF is dated 9 October. That is harmless, but align it.

5 Should fix and cosmetic

S1. Quote the CC14 coassociativity axiom verbatim (no twist). Demote step (3) to a remark. Change “flip and coassociativity” to “universality + lift-coassociativity” in the headline sentences (§2, item 4).

S2. Thm 13 step (1): derive v from $T(+) \circ \langle \ell\pi_1, 0_T\pi_2 \rangle$ and cite $\ell \circ 0 = 0_T \circ 0$ for $\delta(1) = 1 \otimes 1$.

S3. Thm 18 closing and Rem. 23: cite Thm 13 step (1). Do not identify the lift with W ’s.

- S4.** Novelty tone-down, the second half of my old N6. “The one new idea” (l. 142), “not only faithful but cleaner” (l. 38), “strictly outside the Lanfranchi–Lemay hypotheses” (l. 382–383), and §8’s “strictly weaker reasons . . . isolating exactly which uses of ring structure in [9] were essential (none . . .)”. Freeness here is imposed by working in $\text{FreeMod}_{\mathbb{N}}$, and the direct-sum rank count works just as well for free modules over a ring. Lanfranchi–Lemay need the PID because their category contains non-free projectives. Yours does not, by fiat. Call it “a rig-internal port”, not evidence that ring structure was inessential.
- S5.** N4 residues. Replace the proof of Lemma 9 with the atoms argument, which is already in Thm 13 step (2). Demote Lemma 7 to a remark. The counts need only $\text{rank}(M \otimes M) = n^2$ in modules.

Cosmetic: C1–C7 from the table. Also: step (3)’s “no exotic non-representable flip to escape through” is unclear; Rem. 3’s “the substitution is the unit of the adjunction between $(-) \otimes W$ and the Kähler functor” is unsupported; delete both or justify them.

Not checked by me: the Lean development, and Lanfranchi–Lemay’s own definition of “solid” and their Ex. 4.20–4.21 (cited in Rem. 15 and Table 1).

6 Recommended registry grades

Claim	Grade	Note
Thm 22, module form $((-) \otimes D$ on $\text{FreeMod}_{\mathbb{N}}$, any rank: exactly y, W)	proved	Thm 13 is now in the text. S1–S3 are one-line inputs, and I checked each
Thm 11 / Thm 18 (finite rank)	proved	S3 for the ℓ_M clause
$\mathbb{N} \cdot y$ carries no tangent structure	proved	coassociativity + universality
“ $\mathbb{N} \cdot y$ is solid”	definitional	true only as “abstractly solid” (M3)
Rem. 14, $\leftarrow$ monoidal flip route	proved (under its hypotheses)	brute-force check for $ K \leq 3$
“ ∂ only nontrivial . . . on $\text{Lin}(\text{SPoly})$ ” (scoped)	checked-sober	honest as “restriction of the ∂ -structure”
§1/§8/§9 unscoped “only nontrivial way”	overclaim	M1

Overall: accept after minor revisions. The theorem is proved, and the write-up needs M1–M4 before the 13th.

Safe to say at Strathclyde

“On free $\mathbb{N}$ -modules, of any rank, the representable first-order tangent structures $(-) \otimes D$ are exactly the trivial one and the dual numbers W . Universality makes the vertical lift an isomorphism. Over $\mathbb{N}$, an isomorphism permutes bases, and lift-coassociativity then forces the diagonal, so the rank is at most 1. W is the restriction of the container-derivative structure on SPoly to its linear maps.” Do not say “ $\mathbb{N} \cdot y$ is solid” without “abstractly”. Do not say that differentiation is the only nontrivial tangent structure on polynomial functors.